

Lecture 10: Content Moderation

HUM 4441: Engineering Ethics for Computer Science Engineers

Reaz Hassan Joarder, Samnun Azfar, and Yousuf Ibne Kamal Niloy

Junior Lecturer

{reazhassan, samnunazfar, yousufniloy}@iut-dhaka.edu

Department of CSE

Islamic University of Technology

AI Recommender Systems

The Chief Means of Content Moderation

Recommender systems **decide what you see**. They are the primary mechanism by which platforms moderate content today - not a separate add-on system.

- Social media recommenders (“algorithms”)
- News media recommenders (“click bait”)
- Retail sites, search engines, GPTs



AI Recommender Systems: Widely Adopted

One of the most **effective** and **widely-adopted** AI applications - a major force in business, marketing, and news media.

Many Successful Techniques

- Matrix factorization
- Bayesian classifiers and decision trees
- Collaborative filtering, k nearest neighbors
- Recurrent neural networks and transformers, etc.

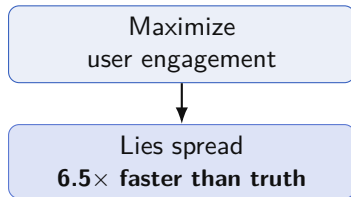


Engagement Maximization: The Downside

AI recommenders are trained to **maximize engagement**. The result is the worldwide spread of:

- Lies, slander, hate speech, harmful misinformation
- Sensational news coverage

This drives **calls for content moderation** - to avoid harmful side effects of profit-maximizing content algorithms.



Factoid: Lies spread 6.5 times faster on social media than truth.

Forms of Content Moderation

1 Take Down

Removing material entirely from the platform.

2 Demote

Putting material at the end of the recommended list - in effect, the same as taking it down.

3 Flag

Flagging material as false or offensive, often alongside reliable sources.

Recommender systems can implement **all** of these forms of moderation.

Content Moderation: An Ad Hoc Landscape

Current Policies Are Ad Hoc

- Respond to complaints
- Change with ownership and political climate
- We need a **principled approach**

Rather than resolve all the issues, this lecture focuses on **two case studies**: inciting violence on YouTube, and social media's impact on young people.

Note: this is about **voluntary** platform policy, not government regulation.

Case Study 1

Inciting Violence on YouTube

Case Study: Inciting Violence - Background

A very high-profile dilemma that raised issues **not resolved to this day**.

The Film

An amateurish film, *Innocence of Muslims*, was uploaded to YouTube on 1 July 2012.

- Packed with **lies** and misinformation
- Highly **offensive** due to negative portrayal of Islam



Innocence of Muslims

2012 · Drama/Short · 14 mins



Wikipedia

https://en.wikipedia.org/wiki/Innocence_of_Muslims

Innocence of Muslims

Innocence of Muslims is a 2012 anti-Islamic short film written and directed by Nakoula Basseley Nakoula under the pseudonym Sam Bacile. ... Two versions of the 14 ... [Read more](#)

Plot and description

Production

Screening and Internet upload

Case Study: Inciting Violence - Reaction

- Protests worldwide, some **violent**
- Reportedly **50 deaths**, mainly in Pakistan
- Most protests anti-U.S., because the film maker lived in the U.S.
- U.S. government did **not** ban the video



Case Study: Inciting Violence - Google's Response

No Legal Authority

President Barack Obama asked Google (owner of YouTube) to take down the film - but he had **no legal authority** to require it.

Google Refused

Google refused to remove the video globally, but **removed it in some countries.**



Case Study: Inciting Violence - Google's Position

Stated Rationale

The post was consistent with “company policy.”

“It is against the Islam religion [sic] but not Muslim people.”

As reported in the NY Times, 14 Sep 2012.

The Open Question

YouTube now has a laundry list of “community guidelines” - but what is the underlying **principle**?



Generalizability: Is Allowing Lies Deceptive?

- Does merely posting lies cause people to believe something YouTube knows is false?
- Only if users regard the video's appearance as a claim or endorsement **by YouTube**
- But users know YouTube hosts all sorts of contradictory views - so there is no obvious endorsement

Conclusion

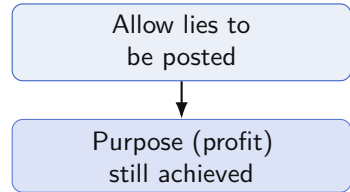
It is hard to make the case that YouTube (as opposed to those who upload the lies) is **deceiving** people.

Generalizability: Would the Policy Still Work?

Would YouTube continue to achieve its purpose even if **all** online sites allowed lies to be posted?

Probably Yes

YouTube's purpose is to **make money**, not to convince anyone to believe its content. Practically all online sites already allow lies, and YouTube still makes tons of money.



Generalizability: Is Removal Generalizable?

Removing False Videos

Removing false videos **because they reduce utility** may **not** be generalizable - generalized private censorship of information may destroy more utility than it creates.

Removing Incendiary Videos

Removing incendiary videos because they **may incite violence** is generalizable - removal would still reduce the probability of violence even as a general practice.

Utilitarian Analysis: Arguments Against Moderation

Argument 1: Causal Distance

"YouTube didn't kill anyone in a riot - the rioters did." **Response:** for the utilitarian principle, only the *consequences* of YouTube's policy matter - it doesn't matter that other people's conduct is involved.

Argument 2: "Ought Implies Can"

Content moderation is a massive task: YouTube gets 500 hours of video/minute; Facebook has 3 billion daily logins. It is **impossible** to screen everything.

Response to Argument 2

- This is a **factual** claim, not an ethical one
- Sites already use AI-based content moderation to direct relevant content to users
- AI can flag questionable posts for human review; users also request takedowns

Scale of Existing Moderation

- YouTube: **10,000** content moderators
- Facebook: **15,000** of 180,000+ employees
- About **100,000** content moderators worldwide

Utilitarian Argument *For* Content Moderation

The Argument

Moderation **increases utility**. Even if it's impossible to catch all harmful posts, it's possible to catch many of them. Failure to do so is a clear violation of the utilitarian principle.

Rebuttal and Counter-rebuttal

"We already do all we can - e.g., we removed all COVID vaccine misinformation."

If so, great - then you agree you **should** do so. And existing recommender technology can clearly be redirected toward this end.

Argument 3: Free Speech

Original Claim

Content moderation violates the **First Amendment rights** of users.

Response: the First Amendment restricts only **government** action - YouTube is a **private company**. This is a generalization argument.

Revised Argument

Content moderation **restricts free speech**, and this is harmful to society.

Question: is it a restriction - or a refusal to give one a particular megaphone?

Autonomy: The Most Demanding Principle

Malicious rumors, terrorist posts, and medical misinformation can lead to death and injury - e.g., in riots. Isn't this a violation of autonomy?

Applying the Principle

A post should be removed when one is **rationally constrained to believe** debilitating harm will result - regardless of who immediately causes the harm.

Implied Consent?

Rioters voluntarily assumed the risk of joining a riot - no autonomy violation for them. But if **innocent bystanders** were hurt, an autonomy violation **does** occur.

Autonomy: How Demanding Can It Get?

Perhaps YouTube managers are rationally constrained to believe that the very **existence** of YouTube will, at some point, interfere with ethical action plans without implied consent - despite best efforts to remove offensive videos.

The Bar for Staying in Business

To operate ethically, moderation must be thorough enough that it is **not irrational** for managers to believe an autonomy violation will never happen.

Take Down Misinformation in General?

- **Not required** by the autonomy principle
- **Could** be justified by the utilitarian principle
- But could also be **ungeneralizable**

A Middle-Ground Option

Flag what fact checkers see as misinformation while providing reliable sources. This does not stifle free speech - and may even promote it - while satisfying the utilitarian principle.

Case Study 1: Summing Up

YouTube must shut down unless it adopts a moderation policy for which managers can rationally believe autonomy violations are not inevitable



A strong **utilitarian imperative** exists to identify such a policy, given YouTube's many benefits



A policy of removing false claims must be carefully crafted to avoid violating **generalizability**



Compromise: flag clearly false claims while providing reliable sources

Case Study 2

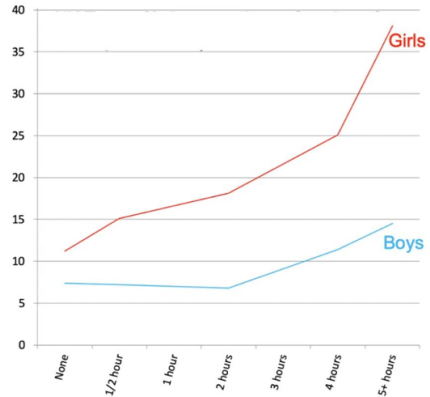
Impact of Social Media on Young People

Impact on Young People: The Data

Depression, Anxiety, and Suicide

All are rising among young people in some countries - roughly coinciding with the rise of social media use.

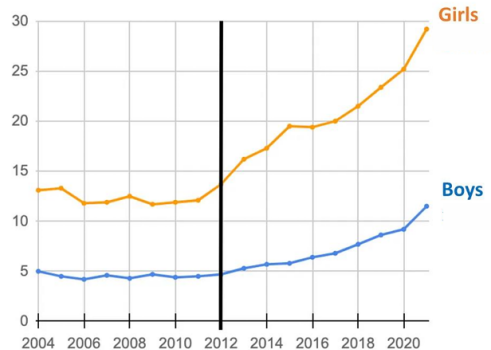
Percent of UK Teens Depressed as a Function of Hours per Weekday on Social Media



Impact on Young People: U.S. Trends

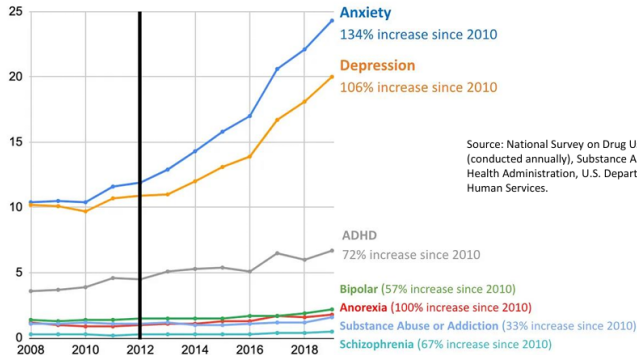
Depression, anxiety, and suicide are all rising among young people in some countries - roughly coinciding with the rise of social media use.

% US Teens with Major Depression



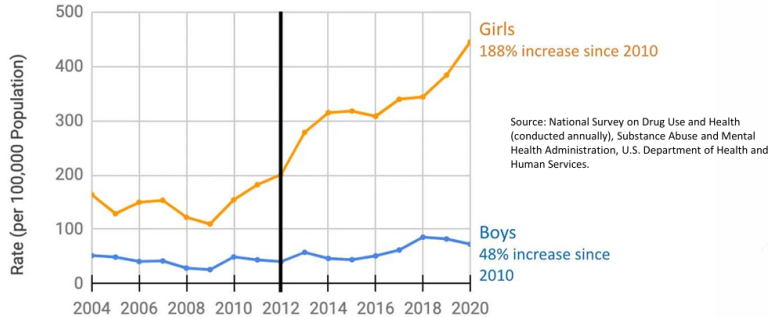
Impact on Young People: College Students

% of US Undergraduates Diagnosed with a Mental Illness



Impact on Young People: Self-Harm Hospitalizations

US Teens Admitted to Hospitals for Nonfatal Self-harm (Ages 10-14)



Alternative Interpretations of the Data

- Heavy social media use may be the **result** of depression rather than its cause
- Rising depression may reflect more frequent **reporting** and acknowledgment of psychological problems
- Rising depression may be real but have **other causes** that happen to coincide with social media use

The Ethical Question That Remains

We can't resolve the factual issues. But we **can** ask: assuming social media overuse causes depression, what should companies do about it?

Autonomy Issues and Children (I)

1 Denying Access

Denying access to YouTube (child or adult) is **no violation of autonomy**. Users cannot have an action plan of being granted access - only the platform can decide whether to grant it.

2 Children Are Not Fully Autonomous

Children often cannot form a coherent rationale for their behavior. In such cases, **parents can forbid** the behavior without violating autonomy.

Autonomy Issues and Children (II)

Children Are Nonetheless Agents

Children sometimes act autonomously - so injuring a child violates autonomy (as well as the utilitarian principle).

Why This Matters

Utilitarian benefits of allowing children online can **never outweigh** autonomy violations - children are far less capable than adults of giving informed consent to the risk of injury, so the chance of an autonomy violation is greater.

Conclusions (I)

Generalization Principle

- Removing clearly harmful content is generalizable
- Failure to remove it is also generalizable
- A carefully crafted policy of removing lies/falsehoods **may** be generalizable
- Flagging lies is generalizable

Autonomy Principle

Online sites can ethically operate only if one can rationally believe their moderated content will **never violate autonomy** without informed consent. Since children are agents, their autonomy must be protected.

Conclusions (II)

Utilitarian principle: a site should make a concerted effort to find an ethical content moderation policy, rather than shut down

It should **at least flag** false and misleading content, while citing reliable sources

Thank You